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Release of complex imaging reports to patients, do radiologists trust AI to help?

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ABSTRACT

Background: As a result of the 21st Century Cures Act, radiology reports are immediately released to patients. However, these reports are often too complex for the lay patient, potentially leading to stress and anxiety. While solutions such as patient portals or providing radiologist contact information have been proposed in the past, new generative artificial intelligence technologies like ChatGPT and Google Gemini may provide the most accessible and scalable method of simplifying radiology reports for patients. Here, we gather the opinions of radiologists regarding this possibility.
Methods: An eight-question survey was sent out to all diagnostic/interventional radiology attendings and clinical fellows at our large academic medical center.
Results: From our survey ($N = 52$), 52.8 % of respondents agreed/strongly agreed that patients should have immediate access to their radiology reports. Only 9.61 % agreed that radiology reports are understandable by the lay patient. Regarding potential avenues to improve patient comprehension of their radiology reports, using artificial intelligence to simplify reports with a manual check by radiologists garnered the most support/strong support (46.2 %). Support of artificial intelligence generated simplifications dropped to (23.1 %) without a manual check.
Conclusion: Patients are increasingly gaining access to their radiology reports, but reports may be too complex for the lay patient. Eventually, artificial intelligence systems may help simplify radiology reports for patients, but there is currently limited support from radiologists.

Introduction

There has been growing emphasis on improving patient engagement in their healthcare. In 2007, the Joint Commission mandated that healthcare organizations “encourage patients’ active involvement in their own care as a patient safety strategy.”[1,2] Following this, the 2009 Health Information Technology for Economic and Clinical Health (HITECH) Act amended the Privacy Rule of the Health Insurance Portability and Accountability Act (HIPAA), improving patients’ right to receive copies of their medical records.[3] Despite these efforts, significant barriers to patient engagement persisted, as patients often had to request providers to receive copies.[4]

To further enhance patient access to their health information, lesser-known provisions of the 21st Century Cures Act (2016) like its Final Rule required patients have access to all their electronic health information.[5,6] The Information Blocking Provision of the Cures Act further necessitated that patients have immediate access to many components of their electronic health information—like radiology reports—with certain exceptions by April 5, 2021.[5]

Traditionally, radiologists have functioned as “doctor-to-doctor” consultants. Hence, before the Information Blocking Provision went into effect, many practices employed a time-delay in releasing reports to prevent patients from encountering potentially distressing findings before their referring providers could review and discuss the results.[7-9] With the Information Blocking Provision, patients have experienced heightened anxiety and providers have faced increased call volume.[9-11]

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While many radiologists support the concept of communicating more directly with patients to explain findings, they remain constrained by time and workload, exacerbated by physician shortages.[12]

These challenges are not unique to the United States. Globally, patients are increasingly accessing their health information through their electronic health records. On November 1, 2022, patients in the UK gained access to all their medical data via the NHS app. The Society of Radiographers and Royal College of Radiologists issued a joint statement welcoming the change while stating "thought also needs to be given as to how these reports are constructed in the future to ensure clear accurate information is conveyed to clinical colleagues but also with consideration that these will be read by patients."[13]

While other care providers remain the primary audience for imaging reports, patients are now becoming frequent readers. Despite this shift, these reports remain overly complex, with medical jargon posing a significant barrier.[14-16] To address the heterogeneous audience of radiology reports, many solutions have been proposed such as patient portals, hyperlinks on medical terms, and summary statements.[11] Information technologies are seen as key enablers, with researchers suggesting "Radiologists must exploit information technologies to craft and deliver meaningful patient-centered reports."[17]

Advancements in Artificial Intelligence (AI) and Natural Language Processing now offer new avenues to make radiology reports more patient centered. Although many studies have shown that these technologies can simplify radiology reports for patients, the attitudes of radiologists toward their use remain unexplored.[18-20] Hence, in this study, we aim to gather radiologists' opinions on the potential use of AI to simplify radiology reports for patients.

Methods

An eight-question anonymous survey was sent to all 142 clinical fellows and diagnostic/interventional radiology attendings at our large academic medical center via the departmental mailing list through the Qualtrics survey platform. Responses were accepted from April 1 to May 1, 2024. No financial incentives were provided to responders. To preserve anonymity, no demographic information of survey respondents was collected beyond role: fellow vs. attending. Question 1 was yes/no while all others were five-point Likert scale questions. All questions are available in Fig. 1's legend. R (2022) was used to perform Wilcoxon signed-rank tests and Prism (2023) was used for graphical representation. The study was deemed exempt by our Institutional Review Board.

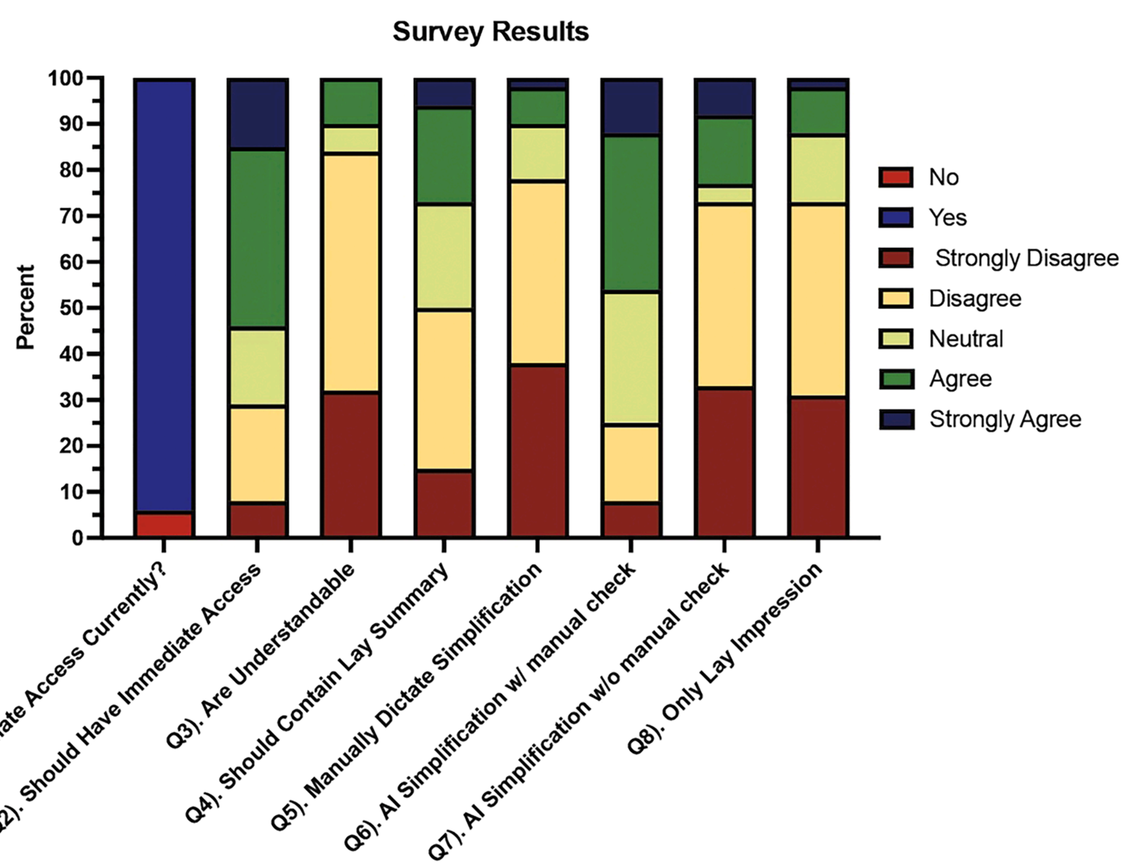

Fig. 1. Survey results depicted as percentages.
Q1). From your understanding, do patients currently receive immediate access to their radiology reports via electronic health records? (yes/no).
Q2). Patients should have immediate access to their radiology reports via electronic health records.
Q3). Imaging reports written by radiologists are understandable by the lay patient (lay patient reads at the 7th/8th grade reading level).
Q4). Imaging reports should contain a summary of the findings and impression in lay language (7th/8th grade reading level).
Q5). Radiologists should manually dictate a simplification of the findings and impression into lay language (7th/8th grade reading level).
Q6). I support artificial intelligence that automatically simplifies the findings and impression into lay language (7th/8th grade reading level), given manual verification (by a radiologist) of the AI generated simplifications. Assume that >85 % of simplified reports contain no inaccurate information while containing all relevant/actionable information and that <1 % contain incorrect information that may harm the patient.
Q7). I support artificial intelligence that automatically simplifies the findings and impressions into lay language (7th/8th grade reading level), given no manual verification (by a radiologist) of the AI generated simplifications. Assume that >85 % of simplified reports contain no inaccurate information while containing all relevant/actionable information and that <1 % contain incorrect information that may harm the patient.
Q8). If there was only going to be one impression on the report, I would support an impression that is purely in lay language (7th/8th grade reading level).

Results

The response rate was 40.7 % (48/118) for attendings and 16.7 % (4/24) for clinical fellows, resulting in a 36.6 % (52/142) overall response rate. All surveys were complete with no missing values. 94.2 % of respondents acknowledged that patients currently receive immediate access to their radiology reports through the Cures Act (Q1, Fig. 1). However, only 53.8 % (28/52) agreed/strongly agreed that patients should receive immediate access (Q2, Fig. 1). No respondents strongly agreed and only 9.61 % (5/52) agreed that radiology reports are understandable by the lay patient (Q3, Fig. 1).

While 26.9 % (14/52) of respondents agreed/strongly agreed (and another 23.1 % (12/52) were neutral)) that radiology reports should contain a lay summary (Q4), only 9.6 % (5/52) agreed/strongly agreed (and another 11.5 % (6/62) were neutral) that radiologists should manually dictate a simplification (Q5, Fig. 1). These differences were significant ($P < 0.0001$). However, significantly more respondents (46.2 % (24/52)) supported/strongly supported AI generated simplifications with a manual check (Q6) compared to manually dictated simplifications (Q5, Fig. 1) ($P < 0.0001$). Support/strong support for AI generated simplifications significantly dropped to 23.1 % (12/52) with no manual check (Q7) compared to with a manual check (Q6) ($P < 0.0001$). Differences between responses for AI generated simplifications without radiologist verification (Q7) and radiologist dictated simplifications (Q5) were not significant ($P = 0.15$). If radiology reports were to contain a single impression, only 11.5 % (6/52) of respondents supported/strongly supported lay impressions (Q8, Fig. 1).

Discussion

As radiology reports have become increasingly patient facing, they remain overly complex for patients.[11,18] In 1992, the Mammography Quality Standards Act initiated a requirement for lay language letters to be sent to patients for mammography results, greatly improving patient access. Even within these lay letters, decreasing the reading level has recently been shown to significantly improve patient follow-up.[21]

Requiring lay language summaries for all reports without a standardized classification system (like BI-RADS is used with breast imaging) would lead to an onerous burden for radiologists. However, AI systems – large language models (LLMs)—have been shown to accurately simplify radiology report impressions.[18,19,22,23] LLMs can produce incorrect information, particularly when asked to make diagnostic decisions.[24] But, the likelihood of errors appears to be diminished when LLMs are purely used to simplify existing information in radiology reports.[19,22,23]

In our survey, nearly all respondents recognized that patients receive immediate access to their reports, but only 53.8 % of radiologists agreed/strongly agreed that this should be the case. Half of this number (26.9 %) agreed/strongly agreed that there should be a simplification. While very few radiologists supported manually dictating a simplification (9.6 %) or AI generated simplifications without a manual check (23.1 %), 46.2 % supported AI generated simplifications with a radiologist check. Unexpectedly, this 46.2 % is significantly greater than the 26.9 % who supported a simplification in the first place ($P = 0.002$), but this may be due to the order of our survey with respondents not realizing the potential of AI until later. At the same time, this 46.2 % is similar to the 53.8 % that support immediate release ($P = 0.77$). The use of the paired Wilcoxon signed-rank test suggests that those who do not support AI with a manual check are largely the same respondents who do not support immediate release in the first place.

When gauging radiologist support for AI generated simplifications, we asked responders to "assume that >85 % of simplified reports contain no inaccurate information while containing all relevant/actionable information and that <1 % contain incorrect information that may harm the patient" based on prior findings.[19] In this previous study, two radiology attendings rating 150 simplified radiology reports across multiple AI models strongly agreed that simplified output contained both no inaccurate information (statement 1) and all relevant and/or actionable information (statement 2) as follows: 86 % (129/150) for ChatGPT-3.5, 83.3 % (125/150) for GPT-4, 75.3 % (113/150) for Google Bard, and 83.3 % (125/150) for Bing.[19] Furthermore, for the criterion of accuracy (statement 1), neutral or worse ratings were recorded as follows: 0 instances for ChatGPT-3.5, 1 for GPT-4, 2 for Google Bard, and 0 for Bing.[19] For the comprehensiveness criterion (statement 2), neutral or worse ratings were observed in 0 instances for ChatGPT-3.5, 0 for GPT-4, 2 for Google Bard, and 0 for Bing.[19]

The high level of accuracy of LLMs in this task assumes manual verification by a radiologist would require few edits. However, before employing these additional requirements on radiologists, additional workflow studies must be conducted, as nearly half of radiologists report burnout.[25,26] Given significant responsibilities already, there is limited incentive for providers to add components to their workflow. These factors may be why few radiologists support simplification, whether with AI or without. Further, these factors may help explain the wide range of responses to each question.

Given limited incentives for radiologists to add components to their workflow, any patient centered solution such as AI generated simplifications may come from legislation – similar to the Mammography Quality Standards Act. Eventually, AI workflow improvements such as report generation may offset the additional burdens of generating patient-centered reports. However, before any AI model is deployed, it must be studied longitudinaly and assessed for implicit biases.[27,28]

Our study cannot determine if AI simplifications are the best available solution, as we do not gather radiologist attitudes towards other methods of making reports more patient friendly like including radiologist phone numbers, explanatory material in patient portals, hyperlinks to common terms, or others. Our study also does not analyze patient attitudes towards AI generated simplifications; however, prior studies already suggest patients want patient friendly reporting.[14] Further, our study is limited as it only gathers responses from a single institution. However, our institution represents a large academic medical system with radiologists reading reports from numerous hospitals, a VA medical center, and outpatient clinics. Despite these limitations, this study is the first to gauge radiologists' attitudes toward using LLMs to simplify radiology reports even though many studies are already exploring the technical feasibility and accuracy of this possibility.[18,19] The medical landscape and care continuum will inevitably change due to new technologies, but how these technologies are implemented remains to be seen.

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